Cascaded Memory Network for Optical Remote Sensing Imagery Cloud Removal

Jiao Liu, Bin Pan, *Member, IEEE*, and Zhenwei Shi, *Senior Member, IEEE*

***Abstract*— Cloud removal is an inevitable task in optical remote sensing images, which aims at restoring high-quality images from cloud-contaminated images. In recent years, deep learning-based image cloud removal methods have utilized convolution neural networks to obtain clean images. However, due to the limitations of the convolution operator, these methods cannot effectively leverage the local and global information of the image. In this article, we propose a cascaded memory network (CMNet) for optical remote sensing imagery cloud removal. The CMNet recycles previously captured information to form a memory mechanism, which is composed of two cascaded components: local information memory module (LIMM) and global information auxiliary module (GIAM). The LIMM aims to obtain local spatial details information of the image via two subnetworks, and the GIAM tries to further restore the detail of the image from a global perspective. In the LIMM, two subnetworks are constructed to capture the details from coarse to fine, each of which includes continuous memory descriptors that describe the local details of the image and hierarchical memory correlation descriptors that adaptively integrate relevant features. In the GIAM, we design the Swin cloud remove transformer layer and explore an adaptive normalization (AdaNorm) to cope with unevenly distributed thin clouds, and further provide theoretical proof for the existence of the required solution. Experimental results indicate that our method can remove clouds while maintaining the detailed information of the image. The code is available at https://github.com/Lab-PANbin/.**

***Index Terms*— Cascaded memory network (CMNet), cloud removal, remote sensing images.**

I. Introduction

OPTICAL remote sensing imagery is becoming more accessible with the development of Earth observation technologies. However, these optical remote sensing images are vulnerable to cloud cover [1], [2], [3], [4], which leads to the loss of some important information and has a critical influence on downstream tasks [5], such as classification [6], [7], [8], [9], object detection [10], [11], [12], semantic segmentation [13], [14], and image retrieval [15]. Therefore, how to remove clouds from images and recover land surface information obscured by clouds has attracted widespread concerns. Generally, the clouds can be roughly divided into two types: thick clouds and thin clouds. Since thick clouds block the signal from the land, while thin clouds can receive part of the signal from the ground, the methods of removing thin clouds and thick clouds are also different.

Thick cloud removal methods typically rely on multimodal or multitemporal data [16]. The multitemporal approaches attempt to utilize images of the same area at different times to provide auxiliary information for the removal of thick clouds. Chen et al. [17] proposed a low-rank sparse decomposition method for blind cloud and shadow removal in multitemporal remote sensing images. Xia and Jia [18] leveraged multitemporal dictionary learning to reconstruct missing information contaminated by thick clouds. References [19], [20], [21], and [22] applied deep learning technology to multitemporal remote sensing images to remove thick clouds. Differing from the multitemporal methods, the multimodal images-based approaches primarily utilize images from different sensors to supplement information for thick cloud removal. References [23], [24], [25], and [26] integrated optical remote sensing data and SAR data, and designed deep neural networks to remove thick clouds. Although the above methods have yielded promising results in thick cloud removal tasks, the acquisition of multimodal and multitemporal data is limited.

For the task of thin cloud removal, many single image-based cloud removal methods have been developed. Early-period thin cloud removal methods were primarily based on traditional image processing techniques. For instance, [27], [28], and [29] designed a homomorphic filter-based method to remove low-frequency cloud noise. Furthermore, [30], [31], and [32] applied dark channel prior [33] in image dehazing to achieve remote sensing image cloud removal. To exploit the spectral information of images, some literatures [34], [35] have designed spectral analysis technology to remove thin clouds. In addition, [36], [37], [38], and [39] leveraged image transformation strategy to remove thin clouds. However, these traditional methods require complex manual designs and are not flexible enough to handle different datasets.

Since deep learning techniques can automatically learn complex mapping relationships from data, deep learning-based thin cloud removal methods have been widely studied. Some researchers designed thin cloud removal models based on generative adversarial networks (GANs) by utilizing unpaired cloud and cloud-free images. Singh and Komodakis [40]

Manuscript received 31 October 2023; revised 22 February 2024; accepted 7 March 2024. Date of publication 18 March 2024; date of current version 21 March 2024. This work was supported in part by the National Key Research and Development Program of China under Grant 2022YFA1003800 and Grant 2022ZD0160401; in part by the National Natural Science Foundation of China under Grant 62001251 and Grant 62125102; and in part by the Beijing–Tianjin–Hebei Basic Research Cooperation Project under Grant F2021203109. *(Corresponding author: Bin Pan.)*

Jiao Liu is with the College of Computer Science, TKLNDST, Nankai University, Tianjin 300350, China (e-mail: jiaoliu@mail.nankai.edu.cn).

Bin Pan is with the School of Statistics and Data Science, KLMDASR, LEBPS, and LPMC, Nankai University, Tianjin 300350, China (e-mail: panbin@nankai.edu.cn).

Zhenwei Shi is with the Image Processing Center, School of Astronautics, Beihang University, Beijing 100191, China (e-mail: shizhenwei@buaa.edu.cn).

Digital Object Identifier 10.1109/TGRS.2024.3376609

1558-0644 © 2024 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 03:30:23 UTC from IEEE Xplore. Restrictions apply.

introduced a Cloud-GAN with unpaired images for removing thin clouds. Li et al. [41] first utilized GAN to generate cloud-free images, and then employed a physical model to reconstruct the original cloud-contaminated images. Zi et al. [42] designed a physical-model-based CycleGAN method that does not require training data to be paired. Other methods developed deep learning-based thin cloud removal models using pairs of cloud images and cloud-free images. For instance, some researchers have designed thin cloud removal models based on convolutional neural networks. Li et al. [43] proposed a deep residual symmetrical concatenation network based on encoding–decoding architecture to preserve detailed information about the image. Zhou et al. [44] proposed a multiscale attention residual network for cloud removal. Wen et al. [45] introduced a residual channel attention network for thin cloud removal. Yu et al. [46] proposed a multiscale distortion-aware thin cloud removal network. Zi et al. [47] introduced a wavelet-integrated convolutional neural network for thin cloud removal. Guo et al. [16] proposed fast Fourier convolution for thin cloud removal. In particular, the GAN can be applied not only to the task of thin cloud removal from unpaired images but can also be employed to design a thin cloud removal model based on paired data. For example, Enomoto et al. [48] designed a multispectral conditional GANs for cloud removal by extending conditional GANs [49]. Wen et al. [50] utilized the Wasserstein GAN to remove thin cloud in YUV color space. Xu et al. [1] designed a multiscale generative adversarial net for cloud removal. In addition, some literatures [51], [52] introduced various attention mechanisms into GANs to obtain high-quality cloud-free images. Different from the previous method of removing clouds using the adversarial network in the generative model, Ding et al. [53] introduced a cloud removal method based on conditional variational autoencoders. Jing et al. [54] designed cloud removal methods based on the diffusion model.

However, most of the existing deep learning-based thin cloud removal methods utilize convolution operators to reconstruct images. On the one hand, they do not fully employ the features extracted by the convolution structure. On the other hand, they are limited by the locality of the convolution operator and cannot capture the dependence of long-range pixels, resulting in incomplete information in the restored image.

In this article, we design a cascaded memory network (CMNet) for optical remote sensing imagery cloud removal. As a whole, the proposed CMNet is a multistage network that reuses previously captured information to form a memory mechanism. To fully extract the local spatial detail information of the image, the first two stages of CMNet form a local information memory module (LIMM). In LIMM, there are two important components, continuous memory descriptor and hierarchical memory correlation descriptor. The continuous memory descriptor also reuses the previously extracted features to obtain local detailed information about the image. Based on the continuous memory descriptor, the hierarchical memory correlation descriptor adaptively fuses relevant features at different levels. Furthermore, in order to recover detailed information about the image from a global perspective, we design the third stage of CMNet as a global information auxiliary module (GIAM). In GIAM, we designed a Swin cloud remove transformer layer that models long-distance dependencies. Especially, in the Swin cloud remove transformer layer, we explore an adaptive normalization (AdaNorm) to cope with unevenly distributed thin clouds and further provide theoretical proof for the existence of the required solution.

The main contributions of this article can be summarized as follows.

1) We propose a CMNet for optical remote sensing imagery cloud removal, which improves locality while capturing long-distance dependencies through multistage network design and information reuse.
2) We develop a LIMM to recover the local spatial detail information of cloud-contaminated images. In particular, the LIMM not only reuses the previously extracted features via continuous memory descriptors but also adaptively fuses features from different layers using hierarchical memory correlation descriptors.
3) We design the Swin cloud remove transformer layer in the GIAM to efficiently model long-distance pixels. Meanwhile, we provided theoretical proof that the AdaNorm technique in the Swin cloud remove transformer layer is feasible.

II. Methodology

A. Overview

For thin cloud removal tasks, the degradation model [53], [55], [56], [57] can be defined as an additive model $\mathbf{I} = \mathbf{I_c} + \mathbf{I_m}$ or a nonlinear model $\mathbf{I} = (1 - \mathbf{I_m}) \cdot \mathbf{I_c} + \mathbf{I_m}$, where $\mathbf{I}$ denotes the cloud image, $\mathbf{I_c}$ represents the clear image, and $\mathbf{I_m}$ is the thickness map of the thin cloud. In this article, we propose a cascaded network CMNet. The overall architecture of CMNet is shown in Fig. 1, which includes two cascaded components: LIIM and GIAM. It is worthing that the LIIM and the GIAM consist of a three-stage network. The LIIM aims to obtain local spatial details information of the image via two subnetworks. Inspired by [58], [59], and [60], the inputs of these two subnetworks are divided into patches at different levels for specific local learning. In particular, each subnet includes a series of continuous memory descriptors and hierarchical memory correlation descriptors based on convolutional neural networks. Furthermore, we designed the Swin cloud remove transformer layer in the GIAM. By introducing the Swin cloud remove transformer layer, we can effectively model the global dependencies of images and assist the network in restoring detailed information on images.

B. Local Information Memory Module

To restore the local details of the image, we design the LIIM. The LIIM comprises two cascaded subnets, each containing some continuous memory descriptors and several hierarchical memory correlation descriptors.

1) Continuous Memory Descriptor: Considering that dilated convolution can better capture the detailed information of images, inspired by [61], [62], [63], [64], and [65], we designed a continuous memory descriptor based on dilated convolution for image cloud removal. As shown in Fig. 2, the continuous memory descriptor consists of dilated convolution, ReLU activation function, concat module, and residual connection. To build a continuous memory mechanism, we exploit

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 03:30:23 UTC from IEEE Xplore. Restrictions apply.

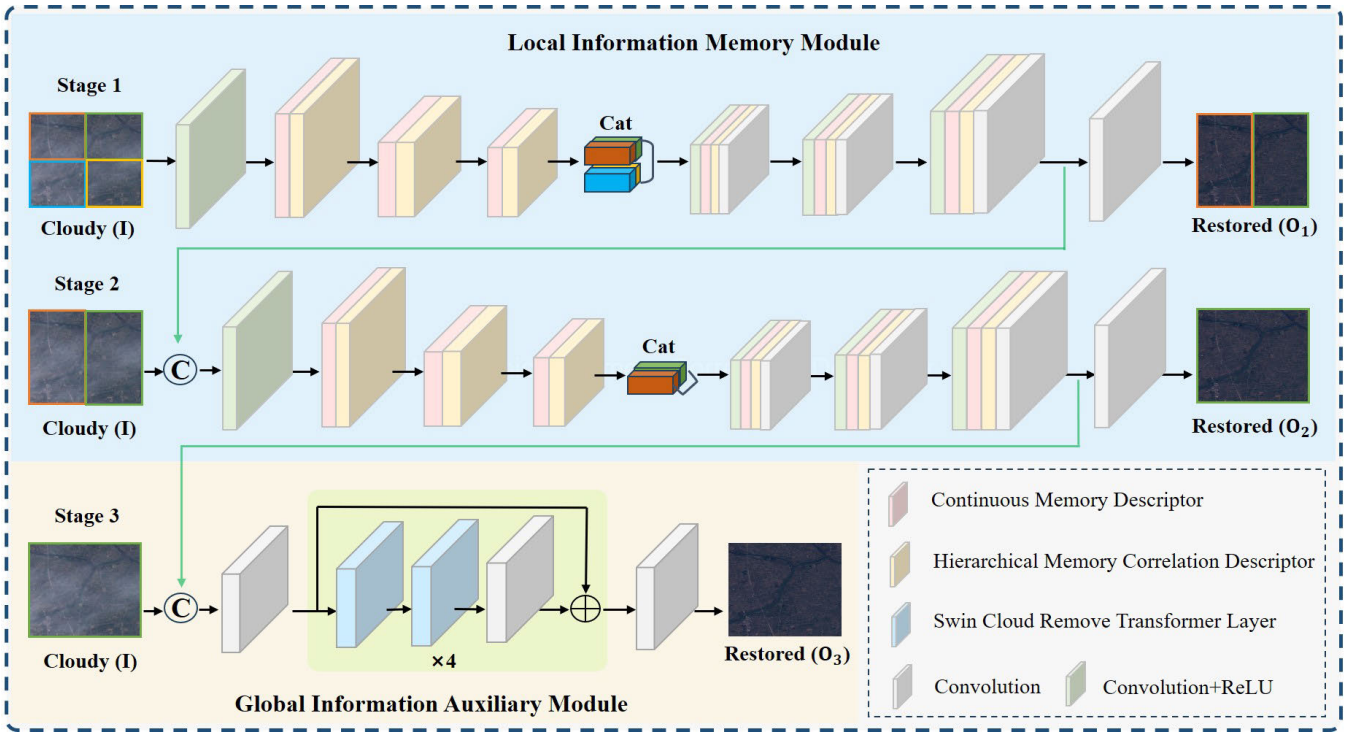


Fig. 1. Overall architecture of the proposed CMNet. The two key components LIMM and the GIAM of CMNet form a multistage network. Each stage takes thin cloud image **I** as input and outputs corresponding cloud removal results $\mathbf{O_1}$, $\mathbf{O_2}$, and $\mathbf{O_3}$. Specifically, the features before the first stage cloud removal result $\mathbf{O_1}$ will be connected to the input of the second stage subnet, and the features before the second stage cloud removal result $\mathbf{O_2}$ will be fed to the third stage. These operations enable the features extracted in the first two stages to be reused while refining the images recovered from the third stage subnet.

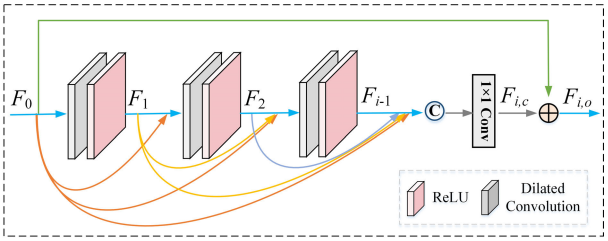


Fig. 2. Architecture of the proposed continuous memory descriptor.

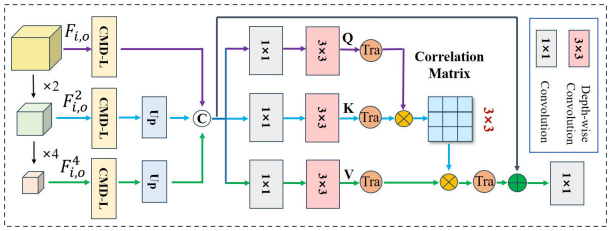


Fig. 3. Architecture of the proposed hierarchical memory correlation descriptor. The continuous memory descriptor without residual connections is called CMD-L.

the output information of the previous layers. Concretely, the input of each dilated convolution is composed of the outputs of previous layers, which can be expressed as follows:

$$F_{i-1} = \mathrm{ReLU}_{i-1}\left(\mathrm{DConv}_{i-1}\left(\left[F_0, F_1, \ldots, F_{i-2}\right]\right)\right) \quad (1)$$

where DConv_{i-1} denotes the dilated convolution of layer $i-1$, ReLU_{i-1} represents the activation function of layer $i-1$, F_{i-2}, and F_{i-1} represent the output of layer $i-2$ and layer $i-1$, respectively. Furthermore, the output feature of each layer is fed into the concat module for local feature fusion. Next, the fused features are fed into a 1×1 convolution to be consistent with the original features. The formulaic description of the above process can be expressed as follows:

$$F_{i,c} = \mathrm{Conv}_{1\times1}\left(\left[F_0, F_1, \ldots, F_{i-2}, F_{i-1}\right]\right) \quad (2)$$

where $\mathrm{Conv}_{1\times1}$ denotes 1×1 convolution. Finally, a residual connection is incorporated into the continuous memory descriptor module to boost the flow of information

$$F_{i,o} = F_0 \oplus F_{i,c} \quad (3)$$

where $\oplus$ represents an addition operation.

2) Hierarchical Memory Correlation Descriptor: To further restore the missing local detail information of the image, the hierarchical memory correlation descriptor attempts to utilize features from different hierarchical levels to complement each other. As shown in Fig. 3, for the feature obtained from the continuous memory descriptor, we utilize bilinear interpolation to generate other hierarchical features, which can be expressed as follows:

$$F^2_{i,o} = \downarrow_2 \mathrm{Bi}\left(F_{i,o}\right) \quad (4)$$

where $\downarrow_2$ represents downsampling $\times2$, Bi denotes bilinear interpolation operation, $F^2_{i,o} \in R^{H/2\times W/2\times C}$, and $F_{i,o} \in R^{H\times W\times C}$. In addition,

$$F^4_{i,o} = \downarrow_4 \mathrm{Bi}\left(F_{i,o}\right) \quad (5)$$

where $\downarrow_4$ denotes downsampling $\times4$, $F^4_{i,o} \in R^{H/4\times W/4\times C}$.

Next, $F_{i,o}$, $F^2_{i,o}$, and $F^4_{i,o}$ are fed into the continuous memory descriptor without residual connections for feature refinement. For the refined different hierarchical features, we seek to further explore the correlations between them. Due to the inconsistent spatial dimensions of the three hierarchical features, we perform upsampling operations on $F_{i,o}$ and $F^2_{i,o}$ to

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 03:30:23 UTC from IEEE Xplore. Restrictions apply.

maintain dimensional consistency. Then, these spatially consistent features are fused, which can be defined as follows:

$$F_{c,o} = \left[F_{i,o}, \uparrow^2 \mathrm{Bi}\left(F_{i,o}^2\right), \uparrow^4 \mathrm{Bi}\left(F_{i,o}^4\right)\right] \tag{6}$$

where $F_{c,o} \in R^{H\times W\times 3C}$, $\uparrow^2$ and $\uparrow^4$ represent upsampling $\times 2$ and $\times 4$, respectively.

Furthermore, we leverage the self-attention mechanism to learn the feature correlations. Similar to [66], we utilize 1×1 convolution to aggregate cross-channel information. To generate $\mathbf{Q}$, $\mathbf{K}$, and $\mathbf{V}$, we employ depth-wise convolutions to simulate the mapping function. This process can be defined as follows:

$$\begin{aligned}\mathbf{Q} &= \mathrm{DW}_{3\times 3}\left(\mathrm{Conv}_1\left(F_{c,o}\right)\right),\\ \mathbf{K} &= \mathrm{DW}_{3\times 3}\left(\mathrm{Conv}_1\left(F_{c,o}\right)\right),\\ \mathbf{V} &= \mathrm{DW}_{3\times 3}\left(\mathrm{Conv}_1\left(F_{c,o}\right)\right)\end{aligned} \tag{7}$$

where Conv_1 denotes 1×1 convolution and $\mathrm{DW}_{3\times 3}$ represents 3×3 depth-wise convolution. Then, we perform dimensional transformations on $\mathbf{Q}$ and $\mathbf{K}$ to facilitate the calculation of feature correlation. The layer correlation matrix is defined as follows:

$$\mathbf{M}_c = \mathrm{softmax}(\mathrm{Tra}(\mathbf{Q}) \otimes \mathrm{Tra}(\mathbf{K})/\alpha) \tag{8}$$

where Tra denotes dimensional transformation operation, α is a scale factor, $\otimes$ denotes matrix multiplication, $\mathbf{M}_c \in R^{3\times 3}$, $\mathrm{Tra}(\mathbf{Q}) \in R^{3\times HWC}$, and $\mathrm{Tra}(\mathbf{K}) \in R^{HWC\times 3}$.

Finally, we multiply the correlation matrix by $\mathbf{V}$ and perform residual operations on it with the original cascaded features. This process is represented as follows:

$$F = \mathrm{Tra}(\mathrm{Tra}(\mathbf{V}) \otimes \mathbf{M}_c) \oplus F_{c,o} \tag{9}$$

where $\mathrm{Tra}(\mathbf{V}) \in R^{HWC\times 3}$.

C. Global Information Auxiliary Module

Furthermore, from the perspective of restoring global context information, we propose the GIAM in the third stage of the model. The GIAM is composed of some convolutional layers and the Swin cloud remove transformer layer. Next, we will provide a detailed introduction to the Swin cloud remove transformer layer.

1) Swin Cloud Remove Transformer Layer: To effectively model the global dependencies of images, inspired by [67], [68], and [69], we design the Swin cloud remove transformer layer. The Swin cloud remove transformer layer is composed of three modules: AdaNorm, multihead self-attention (MSA), and multilayer perception (MLP). For a given input feature, the Swin cloud remove transformer layer and first feeds the input feature into the AdaNorm layer to normalize. Unlike the previous LayerNorm normalized technology, the AdaNorm allows the model to handle thin clouds with different distributions. Let $\mathbf{x} = (x_1, x_2, \ldots, x_d)$ be the input vector, and the dimension of the vector is d. The process of normalization $\mathbf{x}$ utilizing AdaNorm is as follows. First, we calculate the mean and standard deviation of the vector, which can be formulated as follows:

$$\mu = \frac{1}{d}\sum_{j=1}^{d} x_j, \tag{10}$$

$$\sigma = \sqrt{\frac{1}{d}\sum\nolimits_{j=1}^{d}\left(x_j - \mu\right)^2} \tag{11}$$

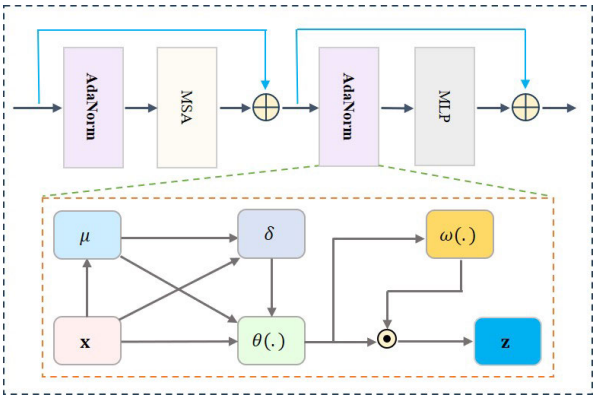


Fig. 4. Architecture of the proposed Swin cloud remove transformer layer.

where μ denotes mean and σ represents standard deviation. Next, we standardize vectors, which are defined as follows:

$$\theta(\mathbf{x}) = \frac{\mathbf{x} - \mu}{\sigma}. \tag{12}$$

Furthermore, we utilize a transformation function to adaptively adjust the standardized vector

$$\mathbf{z} = \omega(\theta(\mathbf{x})) \odot \theta(\mathbf{x}) \tag{13}$$

where $\omega(\theta(\mathbf{x}))$ is a function about $\mathbf{x}$, $\odot$ is a dot product symbol, and $\mathbf{z}$ is the output of AdaNorm.

Next, the output of AdaNorm is fed into the MSA module to model the global dependencies of the image. As shown in Fig. 4, we establish residual connections between the original features and the features output by MSA. Then, we add AdaNorm and an MLP with two fully connected layers for further feature transformations. Especially, the residual connection is also applied to AdaNorm and MLP modules. The formulaic expression of these processes is as follows:

$$\mathbf{X}_n = \mathrm{MSA}(\mathrm{AdaNorm}(\mathbf{X}_m)) \oplus \mathbf{X}_m, \tag{14}$$

$$\mathbf{X}_o = \mathrm{MLP}(\mathrm{AdaNorm}(\mathbf{X}_n)) \oplus \mathbf{X}_n. \tag{15}$$

In the Swin cloud remove transformer layer, it is worth noting that AdaNorm can adaptively adjust scaling weights based on input. However, there are certain requirements for the transformation function $\omega(.)$ in AdaNorm. For instance, the $\omega(.)$ is differentiable, the average of $\omega(\theta(\mathbf{x}))$ is a constant greater than 0, and the average of $\mathbf{z}$ is bounded.

2) Theoretical Proof of the Feasibility of AdaNorm: In this section, we convert the theoretical proof that the AdaNorm technique is feasible into a theoretical proof of the existence of the mapping function $\omega(.)$, based on [69].

Lemma 1: Assume $\omega(\theta(x_j))$ is differentiable, $(1/d)\sum_{i=1}^{d} \omega(\theta(x_j)) = K > 0$, and $(1/d)\sum_{i=1}^{d} z_j < Q (Q > 0)$. Then, there exists a unique optimizer, which can be defined as follows:

$$\omega\left(\theta\left(x_j\right)\right) = K\left(1 - \eta\theta\left(x_j\right)\right). \tag{16}$$

Due to $1 - \eta\theta(x_j) < 0$ will change the direction of the vector, and we hope that $\omega(\theta(x_j)) > 0$ holds. Therefore, the $\theta(x_j) < 1/\eta$ should hold. According to absolute value inequality, the $\theta(x_j) < 1/\eta$ also holds. Furthermore, based on the Chebyshev's Inequality, we can obtain

$$\begin{aligned} P\left(|\theta\left(x_j\right)| < 1/\eta\right) &= P\left(|\theta\left(x_j\right) - E\left(\theta\left(x_j\right)\right)| < 1/\eta\right)\\ &\geq 1 - D_{\theta(\mathbf{x})}/(1/\eta)^2\\ &= 1 - \eta^2 D_{\theta(\mathbf{x})}\end{aligned} \tag{17}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 03:30:23 UTC from IEEE Xplore. Restrictions apply.

where $D_{\theta(\mathbf{x})}$ is the variance of $\theta(\mathbf{x}) = (\theta(x_1), \theta(x_2), \ldots, \theta(x_d))$. In addition, we can utilize the (10)–(12) to calculate the mean and variance of $(\theta(x_1), \theta(x_2), \ldots, \theta(x_d))$. It is very easy to verify that the following formula holds:

$$\begin{aligned}\bar{\theta(\mathbf{x})} &= \frac{1}{d}\sum_{j=1}^{d}\theta\left(x_j\right) \\ &= \frac{1}{d}\sum_{j=1}^{d}\frac{\left(x_j-\mu\right)}{\sigma} \\ &= 0, \end{aligned} \tag{18}$$

$$\begin{aligned} D_{\theta(\mathbf{x})} &= \frac{1}{d}\sum_{j=1}^{d}\frac{\left(x_j-\mu\right)^2}{\sigma^2} \\ &= 1. \end{aligned} \tag{19}$$

Since $D_{\theta(\mathbf{x})} = 1$, if we require $|\theta(x_j)| < 1/\eta$ to hold with a probability higher than 99%, in (17), $\eta = 1/10$. Therefore,

$$\omega\left(\theta\left(x_j\right)\right) = K\left(1 - 1/10\theta\left(x_j\right)\right). \tag{20}$$

Finally,

$$\mathbf{z} = K(1 - 1/10\theta(\mathbf{x})) \odot \theta(\mathbf{x}). \tag{21}$$

D. Loss Function

In this article, we exploit Charbonnier loss [70] and edge loss to jointly optimize the proposed model, which can be defined as follows:

$$\mathcal{L} = \sum_{s=1}^{3}(\mathcal{L}_c + \beta\mathcal{L}_e) \tag{22}$$

where s is the number of stages in the CMNet and $\mathcal{L}_c$ is the Charbonnier loss. The Charbonnier loss can be defined as follows:

$$\mathcal{L}_c = \sqrt{\|\mathbf{O}_s - \mathbf{I}_c\|^2 + \varepsilon^2} \tag{23}$$

where $\mathbf{I}_c$ is the clean image, $\mathbf{O}_s$ is the predicted restored image, and ε^2 is the constant. In this article, we empirically set the ε to 0.001. In addition, in (22), $\mathcal{L}_e$ is the edge loss, which can be expressed as follows:

$$\mathcal{L}_e = \sqrt{\|\Delta(\mathbf{O}_s) - \Delta(\mathbf{I}_c)\|^2 + \varepsilon^2} \tag{24}$$

where Δ is the Laplacian operator. Finally, to balance the Charbonnier loss and edge loss, we set $\beta = 0.5$ in (22) following [71] and [60]. In addition, the pseudocode of our proposed algorithm is shown in Algorithm 1.

Algorithm 1 The Process of the Proposed CMNet

Input: Cloudy image $\mathbf{I}$, batch size, epoch, learning rate

for epoch $= 1 : n$ **do**
  Stage1:
  Divide cloudy image $\mathbf{I}$ into four patches: $\mathbf{I_{11}}$, $\mathbf{I_{12}}$, $\mathbf{I_{13}}$, $\mathbf{I_{14}}$;
  Each image patch first passes through the convolution+ReLU module and then feeds several continuous memory descriptors and hierarchical memory correlation descriptors: E($\mathbf{I_{11}}$), E($\mathbf{I_{12}}$), E($\mathbf{I_{13}}$), E($\mathbf{I_{14}}$);
  Concat deep features: [E($\mathbf{I_{11}}$), E($\mathbf{I_{12}}$))], [E($\mathbf{I_{13}}$), E($\mathbf{I_{14}}$)];
  Pass features through several sequentially connected convolution+ReLU module, continuous memory descriptors, hierarchical memory correlation descriptors, and convolution: D([E($\mathbf{I_{11}}$), E($\mathbf{I_{12}}$)]), D([E($\mathbf{I_{13}}$), E($\mathbf{I_{14}}$)]);
  Feed D([E($\mathbf{I_{11}}$), E($\mathbf{I_{12}}$)]) and D([E($\mathbf{I_{13}}$), E($\mathbf{I_{14}}$)]) to the convolutional layer to obtain a clean image $\mathbf{O_1}$.
  Stage2:
  Divide cloudy image $\mathbf{I}$ into two patches: $\mathbf{I_{21}}$, $\mathbf{I_{22}}$;
  Cascade: D([E($\mathbf{I_{11}}$), E($\mathbf{I_{12}}$)]) and $\mathbf{I_{21}}$, D([E($\mathbf{I_{13}}$), E($\mathbf{I_{14}}$)]) and $\mathbf{I_{22}}$;
  The cascade features $\mathbf{I_{c1}}$ and $\mathbf{I_{c2}}$ are fed to convolution+ReLU module, several sequentially connected continuous memory descriptors and hierarchical memory correlation descriptors: E($\mathbf{I_{c1}}$), E($\mathbf{I_{c2}}$);
  Concat deep features: [E($\mathbf{I_{c1}}$), E($\mathbf{I_{c2}}$)];
  Pass features through several sequentially connected convolution+ReLU module, continuous memory descriptors, hierarchical memory correlation descriptors, and convolution: D([E($\mathbf{I_{c1}}$), E($\mathbf{I_{c2}}$)]);
  Feed D([E($\mathbf{I_{c1}}$), E($\mathbf{I_{c2}}$)]) to the convolutional layer to obtain a clean image $\mathbf{O_2}$.
  Stage3:
  Cascade: D([E($\mathbf{I_{c1}}$), E($\mathbf{I_{c2}}$)]) and $\mathbf{I}$;
  The cascade features $\mathbf{I_{c3}}$ first pass through the convolution module and then feed some sequentially connected swin cloud remove transformer layer and convolution layer: G($\mathbf{I_{c3}}$);
  Feed G($\mathbf{I_{c3}}$) to the convolutional layer to obtain a clean image $\mathbf{O_3}$;
  Update the parameters of the CMNet by Eq.(22);
end for

Output: Restored image $\mathbf{O_3}$, $\mathbf{O_2}$, $\mathbf{O_1}$.

III. Experiments

A. Datasets

To validate the proposed method, we conducted experiments on four datasets, T-CLOUD [53], RICE1 [72], RICE2 [72], and WHUS2-CR [73], respectively. Fig. 5 illustrates the paired samples for different datasets. Details of these datasets are as follows.

1) *T-CLOUD:* The T-CLOUD is a thin cloud dataset, which is composed of 2939 real image pairs acquired by the Landsat 8 satellite. Specifically, the interval between cloud images and corresponding clean images is 16 days. In T-CLOUD, the training and test data are divided into 2351 and 588 pairs, respectively, and the size of each image is 256 × 256.

2) *WHUS2-CR:* The WHUS2-CR dataset is acquired from the Sentinel-2A satellite, and the period of acquisition of image pairs was the revisit time of the satellite (ten days) to a minimum of the difference between cloud and clean images. Since Sentinel-2A is a multispectral imaging satellite, following [45], we choose visible bands to construct true color images. It is worth mentioning that WHUS2-CR is a thin cloud dataset. In our experiments, 3776 pairs of 256 × 256 images were used for training and 944 pairs of 256 × 256 images were employed for testing.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 03:30:23 UTC from IEEE Xplore. Restrictions apply.

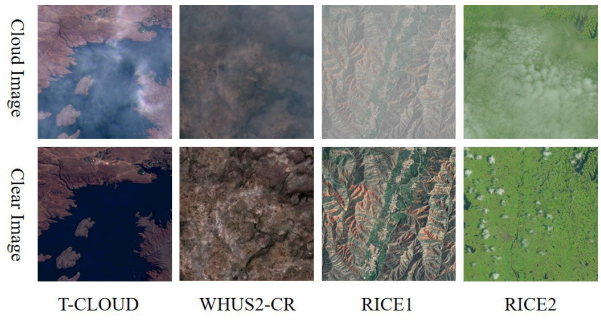


Fig. 5. Sample cloud images from different datasets.

3) *RICE1 and RICE2:* RICE1 and RICE2 are two sub-datasets of the RICE dataset. The size of the images in each dataset is 512 × 512. Specially, the RICE1 is a thin cloud dataset from Google Earth containing 500 image pairs and RICE2 is a thick cloud dataset with 736 image pairs obtained from Landsat 8. Considering the generalization of our algorithm, we also conducted experiments on RICE2 datasets containing thick clouds.

B. Experimental Setup

1) Evaluation Metrics: In this article, we choose both full-reference metrics and no-reference metrics to quantitatively evaluate the performance of the model. For the full-reference, we utilize the peak-signal-to-noise ratio (PSNR) and structural similarity index measure (SSIM), while for the no-reference metric, we chose the natural image quality evaluator (NIQE). It is worth mentioning that larger PSNR and SSIM values, as well as smaller NIQE values, indicate higher image quality.

2) Implementation Detail: We implement the proposed method on the Nvidia GeForce 2080Ti GPU using the Pytorch framework. To optimize the proposed CMNet, we chose an Adam optimizer with an initial learning rate of 0.0002. In addition, we set the batch size to 2 and epoch to 250 to train the proposed CMNet. In particular, in CMNet, we assign the two SCRTL layers to 1 group, for a total of 4 groups, and the number of attention heads is set to 4.

3) Compared Methods: In this article, we choose six cloud removal methods to compare with the proposed methods.

1) DCP [33]: DCP is an image dehazing method based on physical priors, which is widely applied to thin cloud removal from a single image.
2) SPA-GAN [51]: The SPA-GAN is a mono-temporal thin cloud removal method based on a GAN, which introduces a spatial attention mechanism to image cloud removal tasks.
3) CVAE [53]: A probability model based on conditional variational autoencoder for single image thin cloud removal.
4) YUV-GAN [50]: The YUV-GAN utilizes the Wasserstein GAN for single-image thin cloud removal in the YUV color space.
5) RCAN [45]: The RCAN utilizes a residual channel attention mechanism for single-image thin cloud removal.

TABLE I
QUANTITATIVE EXPERIMENTAL RESULTS OF DIFFERENT CLOUD REMOVAL METHODS ON THE T-CLOUD DATASET, WITH THE BEST RESULTS IN RED, SECOND BESTS IN ORANGE, AND THIRD BESTS IN BLUE

Method	PSNR ↑	SSIM ↑	NIQE ↓
DCP	17.5528	0.6786	7.3764
SPA-GAN	27.4199	0.8238	5.3796
YUV-GAN	22.9167	0.8108	6.0647
MS-GAN	28.0514	0.8301	4.9347
RCAN	25.0667	0.8648	4.9937
CVAE	29.6723	0.8940	4.9108
CMNet	31.8955	0.9081	4.9407

TABLE II
QUANTITATIVE EXPERIMENTAL RESULTS OF DIFFERENT CLOUD REMOVAL METHODS ON THE WHUS2-CR DATASET, WITH THE BEST RESULTS IN RED, SECOND BESTS IN ORANGE, AND THIRD BESTS IN BLUE

Method	PSNR ↑	SSIM ↑	NIQE ↓
DCP	15.0854	0.5971	5.4479
SPA-GAN	27.7271	0.8312	5.8353
YUV-GAN	17.7579	0.7646	5.7211
MS-GAN	22.6312	0.7700	5.1884
RCAN	21.3644	0.7598	5.3834
CVAE	29.5068	0.8589	5.8262
CMNet	30.8483	0.9050	4.8191

6) MSGAN [1]: MSGAN is a multiscale generative adversarial net for single image thin cloud removal.

C. Experimental Results

1) Experimental Results on T-CLOUD: Table I shows the quantitative experimental results of CMNet and other comparative methods on the T-CLOUD dataset. From Table I, it can be seen that the CMNet obtains the best results for both PSNR and SSIM metrics. Specifically, the PSNR value obtained by CMNet is 2.2232 dB higher than the second-best CVAE algorithm. For NIQE scores, although our CMNet did not achieve the best results, it is still comparable to the best CVAE algorithm. Furthermore, Fig. 6 shows the results of qualitative experiments on the T-CLOUD dataset with our method and other methods. Obviously, the proposed CMNet can effectively remove the thin clouds while preserving the detailed information of the image. On the contrary, other cloud removal methods cannot effectively remove thin clouds, and on the other hand, the restored image content is too smooth, resulting in the loss of key information in the image. Specifically, the YUV-GAN and DCP methods also exhibit color distortion in certain images.

2) Experimental Results on WHUS2-CR: To further validate the effectiveness of the proposed CMNet, we conducted experiments on the WHUS2-CR dataset. Table II shows the quantitative experimental results of all methods on the WHUS2-CR dataset. From Table II, we can observe that CMNet has achieved the best performance in all evaluation metrics. In addition, Fig. 7 shows the qualitative experimental results of all methods in WHUS2-CR. From Fig. 7, we can observe that the proposed CMNet is still successful in recovering the detailed information on WHUS2-CR. However, some methods generate a certain degree of color distortion on the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 03:30:23 UTC from IEEE Xplore. Restrictions apply.

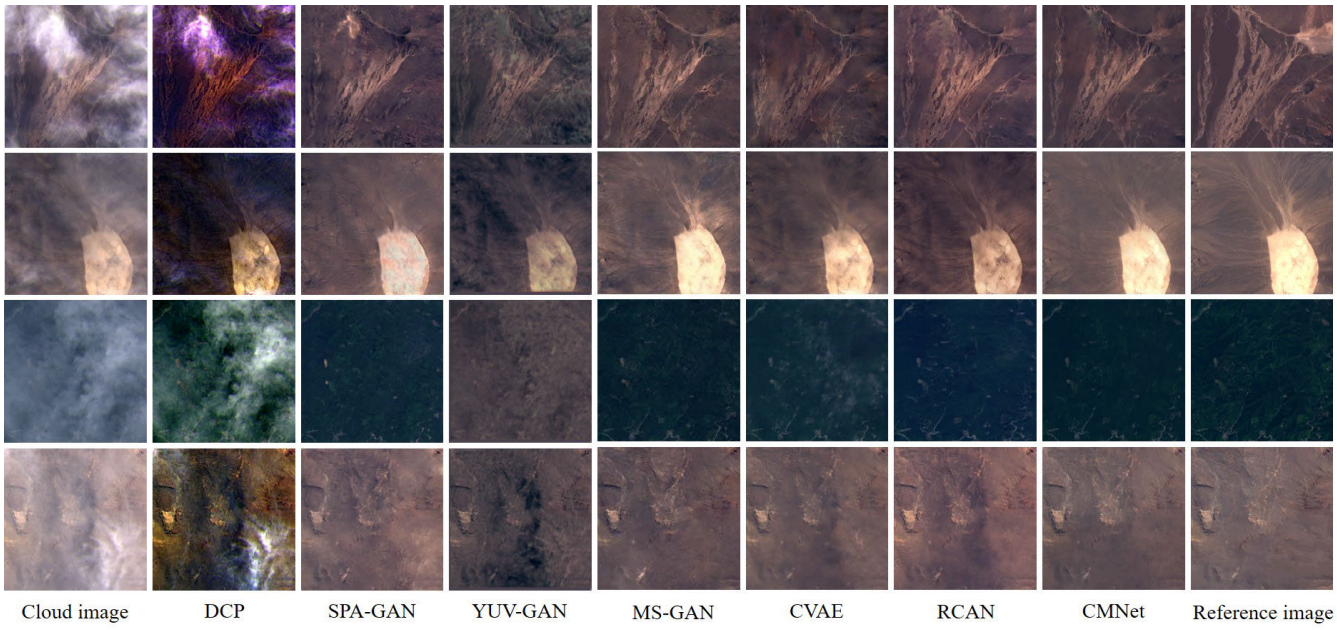


Fig. 6. Qualitative experimental results of different comparison methods on the T-CLOUD dataset. The first and last columns denote the cloud image and reference image. The second column to the penultimate column, respectively, represent different cloud removal methods.

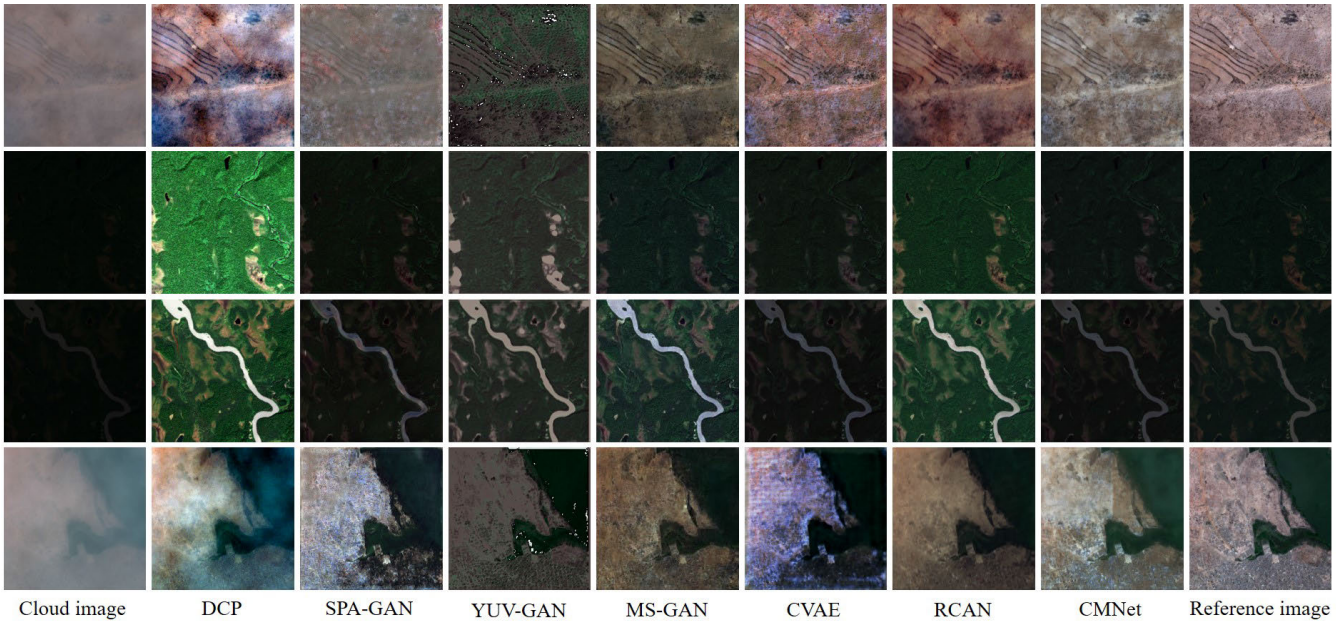


Fig. 7. Qualitative experimental results of different comparison methods on the WHUS2-CR dataset. The first and last columns denote the cloud image and reference image. The second column to the penultimate column, respectively, represent different cloud removal methods.

WHUS2-CR dataset. Compared to other methods, the CMNet yields less color distortion.

3) Experimental Results on RICE1: Similar to other literature, we also conducted experiments on the commonly used thin cloud dataset RICE1. Table III shows the results of the quantitative experiments of all methods on the RICE1 dataset. From Table III, we can observe that the PSNR of CMNet reached 37.5917 dB, the SSIM achieved 0.9696, and the NIQE score ranks second. Fig. 8 illustrates the results of the qualitative experiments on the RICE1 dataset. From Fig. 8, we can see that some methods have better performance in removing certain thin cloud images, but there is still the problem of imperfect cloud removal for images with slightly nonuniform cloud distribution.

4) Experimental Results on RICE2: In particular, we also validate the generalization ability of our method on the thick cloud dataset RICE2. Table IV shows the results of the quantitative experiments of all methods on the RICE2 dataset. As can be seen from Table IV, RICE2 has still achieved better results even on thick cloud datasets. In addition, we also present the quantitative experimental results of the different comparison methods in Fig. 9. Compared to other methods, CMNet can eliminate thick clouds to a certain extent and avoid color distortion while retaining image detail information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 03:30:23 UTC from IEEE Xplore. Restrictions apply.

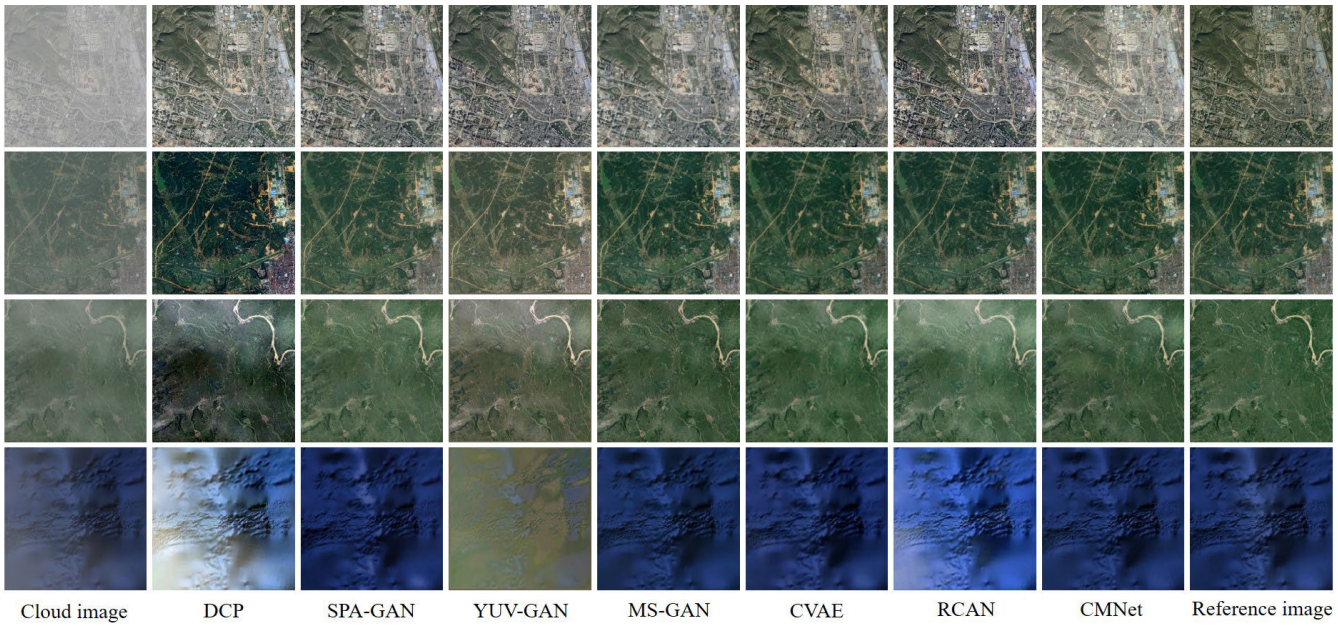


Fig. 8. Qualitative experimental results of different comparison methods on the RICE1 dataset. The first and last columns denote the cloud image and reference image. The second column to the penultimate column, respectively, represent different cloud removal methods.

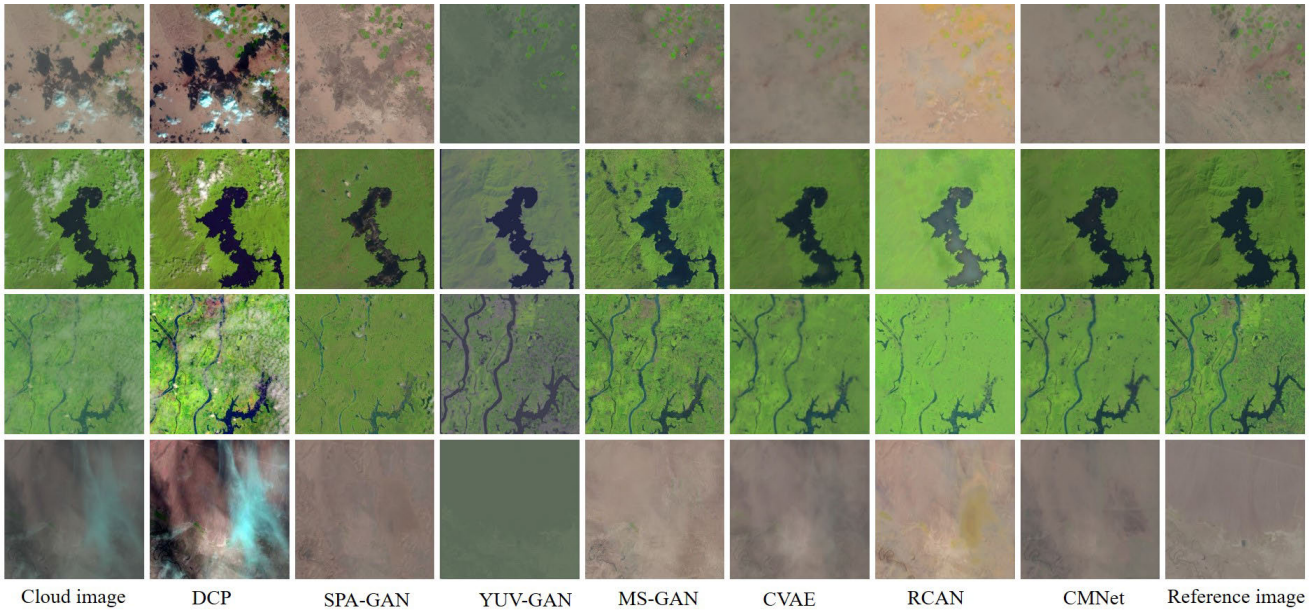


Fig. 9. Qualitative experimental results of different comparison methods on the RICE2 dataset. The first and last columns denote the cloud image and reference image. The second column to the penultimate column, respectively, represent different cloud removal methods.

D. Ablation Study

In this section, we designed a series of ablation experiments to validate the effectiveness of each component of CMNet. We performed ablation experiments on the RICE1 dataset. Figa. 10 and 11 demonstrate the results of the quantitative experiments on RICE1. In Figs. 10 and 11, we can observe that removing the hierarchical memory correlation descriptor not only reduces PSNR and SSIM values but also correspondingly reduces the number of model parameters. In particular, we remove only the feature correlation components in the hierarchical memory correlation descriptor, and the performance of the model decreases accordingly, but the performance is still better than removing the whole hierarchical memory correlation descriptor. In addition, we also explored the necessity of a three-stage network. For example, we removed stage 2 and the GIAM (stage 3) and found that the performance of the model plummeted. Next, we validated the performance of the model after removing only the global information guidance module (stage 3). With the experimental results in Figs. 10 and 11, we can observe that the existence of multiple stages is necessary. By interacting with information from multiple stages, we have achieved complementarity between local and global information. Fig. 12 illustrates the results of qualitative experiments on the ablation of individual components. By comparing

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 03:30:23 UTC from IEEE Xplore. Restrictions apply.

TABLE III

QUANTITATIVE EXPERIMENTAL RESULTS OF DIFFERENT CLOUD REMOVAL METHODS ON THE RICE1 DATASET, WITH THE BEST RESULTS IN RED, SECOND BESTS IN ORANGE, AND THIRD BESTS IN BLUE

Method	PSNR ↑	SSIM ↑	NIQE ↓
DCP	18.7909	0.7690	4.8342
SPA-GAN	29.6012	0.9175	4.9120
YUV-GAN	23.4533	0.8877	5.2224
MS-GAN	32.5041	0.9320	3.8788
RCAN	26.6876	0.9353	4.2707
CVAE	33.0530	0.8856	5.6343
CMNet	37.5917	0.9696	4.1952

TABLE IV

QUANTITATIVE EXPERIMENTAL RESULTS OF DIFFERENT CLOUD REMOVAL METHODS ON THE RICE2 DATASET, WITH THE BEST RESULTS IN RED, SECOND BESTS IN ORANGE, AND THIRD BESTS IN BLUE

Method	PSNR ↑	SSIM ↑	NIQE ↓
DCP	16.9987	0.6450	4.8244
SPA-GAN	30.9325	0.8717	5.0315
YUV-GAN	18.2313	0.8193	7.2918
MS-GAN	32.3450	0.8806	5.3176
RCAN	19.3993	0.8403	3.8781
CVAE	35.5415	0.9185	5.5282
CMNet	37.5605	0.9264	5.2195

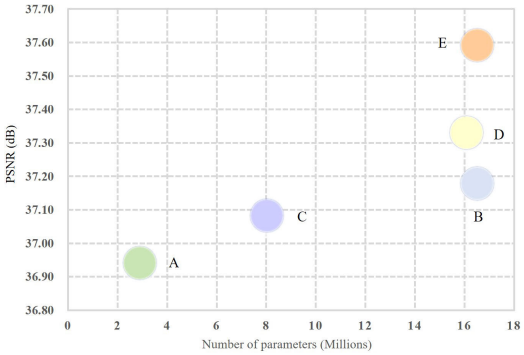


Fig. 10. Ablation quantitative experimental results about PSNR and the total number of parameters on the RICE1 dataset. "A" denotes removing the entire hierarchical memory correlation descriptor. "B" represents deleting feature correlation from the hierarchical memory correlation descriptor. "C" denotes removing stages 2 and the GIAM (stage 3). "D" indicates that there is no GIAM (stage 3) in the CMNet. "E" is our whole model CMNet.

the various subfigures in Fig. 12, we can observe that each of our modules is effective. For example, by comparing (a) and (e), we can observe that hierarchical memory correlation descriptors can further restore the local details of the image. Comparing (d) and (e), we can see that (e) has fewer clouds and the whole image appears to be more coherent. This implies that the GIAM can enhance the effect of cloud removal while preserving the global detail information of the image.

E. Analysis of Model Size

In this section, we analyzed the model sizes of several thin cloud removal methods and conducted experiments on the RICE1 dataset. The experimental results are shown in Table V. From Table V, we can observe that although the SPA-GAN method has a smaller model size, its performance is not as

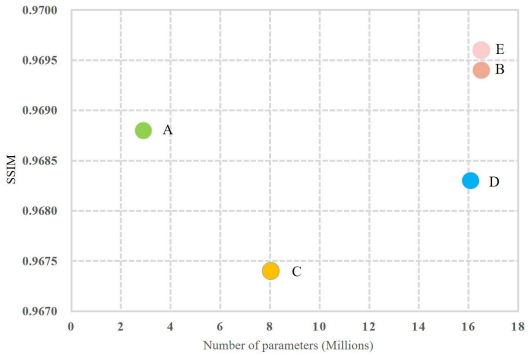


Fig. 11. Ablation quantitative experimental results about SSIM and the total number of parameters on the RICE1 dataset. "A" denotes removing the entire hierarchical memory correlation descriptor. "B" represents deleting feature correlation from the hierarchical memory correlation descriptor. "C" denotes removing stages 2 and the GIAM (stage 3). "D" indicates that there is no GIAM (stage 3) in the CMNet. "E" is our whole model CMNet.

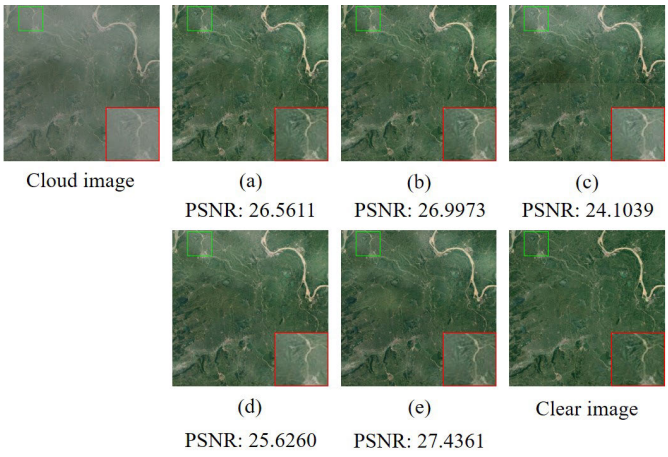


Fig. 12. Ablation qualitative experimental results on the RICE1 dataset. (a) Removing the entire hierarchical memory correlation descriptor. (b) Deleting feature correlation from the hierarchical memory correlation descriptor. (c) Removing stages 2 and the GIAM (stage 3). (d) Indicates that there is no GIAM (stage 3) in the CMNet. (e) Our whole model CMNet. Specifically, the red box content in the image is an amplification of the green box content.

TABLE V

COMPARISON OF PERFORMANCE AND MODEL SIZE OF DIFFERENT THIN CLOUD REMOVAL METHODS

Method	SPA-GAN	YUV-GAN	MS-GAN	RCAN	CVAE	CMNet
PSNR	27.7271	17.7579	22.6312	21.3644	29.5068	30.8483
# Params (M)	2.98	69.05	65.11	3.09	15.42	16.51

good as that of CVAE and our CMNet. Specifically, compared to the CVAE method that achieved the second-best performance, our CMNet model size only increased by 1.09M.

IV. CONCLUSION

In this article, we propose a CMNet for image cloud removal. The CMNet includes the LIMM and the GIAM. The LIMM aims to obtain local spatial details information of the image via two serial subnetworks. The GIAM attempts to restore the detail of the image from a global perspective. We conducted extensive experiments on four datasets, and the experimental results showed that the CMNet can remove clouds while maintaining the detailed information of the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 03:30:23 UTC from IEEE Xplore. Restrictions apply.

image. In the future, we will explore the properties of image cloud removal tasks in a deep way, and customize a more effective framework for image cloud removal.

REFERENCES

[1] Z. Xu, K. Wu, L. Huang, Q. Wang, and P. Ren, "Cloudy image arithmetic: A cloudy scene synthesis paradigm with an application to deep-learning-based thin cloud removal," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5612616.

[2] Z. Zou, W. Li, T. Shi, Z. Shi, and J. Ye, "Generative adversarial training for weakly supervised cloud matting," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 201–210.

[3] F. Xie, M. Shi, Z. Shi, J. Yin, and D. Zhao, "Multilevel cloud detection in remote sensing images based on deep learning," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 10, no. 8, pp. 3631–3640, Aug. 2017.

[4] C. Stubenrauch et al., "Assessment of global cloud datasets from satellites: Project and database initiated by the GEWEX radiation panel," *Bull. Amer. Meteorol. Soc.*, vol. 94, no. 7, pp. 1031–1049, 2013.

[5] T.-Y. Ji, D. Chu, X.-L. Zhao, and D. Hong, "A unified framework of cloud detection and removal based on low-rank and group sparse regularizations for multitemporal multispectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5303015.

[6] J. Zhang, J. Liu, B. Pan, Z. Chen, X. Xu, and Z. Shi, "An open set domain adaptation algorithm via exploring transferability and discriminability for remote sensing image scene classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2021, Art. no. 5609512.

[7] Y. Jia, J. Gao, W. Huang, Y. Yuan, and Q. Wang, "Exploring hard samples in multiview for few-shot remote sensing scene classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5615714.

[8] J. Peng, Y. Huang, W. Sun, N. Chen, Y. Ning, and Q. Du, "Domain adaptation in remote sensing image classification: A survey," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 9842–9859, 2022.

[9] X. Zheng, T. Gong, X. Li, and X. Lu, "Generalized scene classification from small-scale datasets with multitask learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5609311.

[10] X. Dong et al., "Multiscale deformable attention and multilevel features aggregation for remote sensing object detection," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[11] Y. Ye et al., "An adaptive attention fusion mechanism convolutional network for object detection in remote sensing images," *Remote Sens.*, vol. 14, no. 3, p. 516, Jan. 2022.

[12] X. Ye, F. Xiong, J. Lu, H. Zhao, and J. Zhou, "M^2-Net: A multi-scale multi-level feature enhanced network for object detection in optical remote sensing images," in *Proc. Digit. Image Comput., Techn. Appl. (DICTA)*, Dec. 2020, pp. 1–8.

[13] L. Wu, M. Lu, and L. Fang, "Deep covariance alignment for domain adaptive remote sensing image segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5620811.

[14] J. Lu, G. He, H. Dou, Q. Gao, L. Fang, and Y. Deng, "ScoreSeg: Leveraging score-based generative model for self-supervised semantic segmentation of remote sensing," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 16, pp. 8818–8833, 2023.

[15] L. Han, P. Li, A. Plaza, and P. Ren, "Hashing for localization (HfL): A baseline for fast localizing objects in a large-scale scene," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5609916.

[16] Y. Guo, W. He, Y. Xia, and H. Zhang, "Blind single-image-based thin cloud removal using a cloud perception integrated fast Fourier convolutional network," *ISPRS J. Photogramm. Remote Sens.*, vol. 206, pp. 63–86, Dec. 2023.

[17] Y. Chen, W. He, N. Yokoya, and T.-Z. Huang, "Blind cloud and cloud shadow removal of multitemporal images based on total variation regularized low-rank sparsity decomposition," *ISPRS J. Photogramm. Remote Sens.*, vol. 157, pp. 93–107, Nov. 2019.

[18] M. Xia and K. Jia, "Reconstructing missing information of remote sensing data contaminated by large and thick clouds based on an improved multitemporal dictionary learning method," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5605914.

[19] Y. Chen, L. Tang, X. Yang, R. Fan, M. Bilal, and Q. Li, "Thick clouds removal from multitemporal ZY-3 satellite images using deep learning," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 13, pp. 143–153, 2020.

[20] W. Sintarasirikulchai, T. Kasetkasem, T. Isshiki, T. Chanwimaluang, and P. Rakwatin, "A multi-temporal convolutional autoencoder neural network for cloud removal in remote sensing images," in *Proc. 15th Int. Conf. Electr. Eng./Electron., Comput., Telecommun. Inf. Technol.*, Jul. 2018, pp. 360–363.

[21] Y. Chen, Q. Weng, L. Tang, X. Zhang, M. Bilal, and Q. Li, "Thick clouds removing from multitemporal Landsat images using spatiotemporal neural networks," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 4400214.

[22] D. Christopoulos, V. Ntouskos, and K. Karantzalos, "CloudTran: Cloud removal from multitemporal satellite images using axial transformer networks," *Int. Arch. Photogramm., Remote Sens. Spatial Inf. Sci.*, vol. 43, pp. 1125–1132, May 2022.

[23] A. Meraner, P. Ebel, X. X. Zhu, and M. Schmitt, "Cloud removal in Sentinel-2 imagery using a deep residual neural network and SAR-optical data fusion," *ISPRS J. Photogramm. Remote Sens.*, vol. 166, pp. 333–346, Aug. 2020.

[24] W. Li, Y. Li, and J. C. Chan, "Thick cloud removal with optical and SAR imagery via convolutional-mapping-deconvolutional network," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 4, pp. 2865–2879, Apr. 2020.

[25] S. Han, J. Wang, and S. Zhang, "Former-CR: A transformer-based thick cloud removal method with optical and SAR imagery," *Remote Sens.*, vol. 15, no. 5, p. 1196, Feb. 2023.

[26] F. Xu et al., "GLF-CR: SAR-enhanced cloud removal with global–local fusion," *ISPRS J. Photogramm. Remote Sens.*, vol. 192, pp. 268–278, Oct. 2022.

[27] O. R. Mitchell, E. J. Delp, and P. L. Chen, "Filtering to remove cloud cover in satellite imagery," *IEEE Trans. Geosci. Electron.*, vol. GE-15, no. 3, pp. 137–141, Jul. 1977.

[28] J. Liu et al., "Thin cloud removal from single satellite images," *Opt. Exp.*, vol. 22, no. 1, pp. 618–632, Jan. 2014.

[29] H. Shen, H. Li, Y. Qian, L. Zhang, and Q. Yuan, "An effective thin cloud removal procedure for visible remote sensing images," *ISPRS J. Photogramm. Remote Sens.*, vol. 96, pp. 224–235, Oct. 2014.

[30] S. Shi, Y. Zhang, X. Zhou, and J. Cheng, "A novel thin cloud removal method based on multiscale dark channel prior (MDCP)," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[31] J. Li, Q. Hu, and M. Ai, "Haze and thin cloud removal via sphere model improved dark channel prior," *IEEE Geosci. Remote Sens. Lett.*, vol. 16, no. 3, pp. 472–476, Mar. 2019.

[32] X. Pan, F. Xie, Z. Jiang, and J. Yin, "Haze removal for a single remote sensing image based on deformed haze imaging model," *IEEE Signal Process. Lett.*, vol. 22, no. 10, pp. 1806–1810, Oct. 2015.

[33] K. He, J. Sun, and X. Tang, "Single image haze removal using dark channel prior," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 33, no. 12, pp. 2341–2353, Dec. 2010.

[34] M. Xu, X. Jia, and M. Pickering, "Automatic cloud removal for Landsat 8 OLI images using cirrus band," in *Proc. IEEE Geosci. Remote Sens. Symp.*, Jul. 2014, pp. 2511–2514.

[35] B. Zhou and Y. Wang, "A thin-cloud removal approach combining the cirrus band and RTM-based algorithm for Landsat-8 OLI data," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2019, pp. 1434–1437.

[36] G. Hu, X. Li, and D. Liang, "Thin cloud removal from remote sensing images using multidirectional dual tree complex wavelet transform and transfer least square support vector regression," *J. Appl. Remote Sens.*, vol. 9, no. 1, Sep. 2015, Art. no. 095053.

[37] Y. Shen, Y. Wang, H. Lv, and J. Qian, "Removal of thin clouds in Landsat-8 OLI data with independent component analysis," *Remote Sens.*, vol. 7, no. 9, pp. 11481–11500, Sep. 2015.

[38] H. Lv, Y. Wang, and Y. Gao, "Using independent component analysis and estimated thin-cloud reflectance to remove cloud effect on Landsat-8 oli band data," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2018, pp. 915–918.

[39] M. Xu, X. Jia, M. Pickering, and S. Jia, "Thin cloud removal from optical remote sensing images using the noise-adjusted principal components transform," *ISPRS J. Photogramm. Remote Sens.*, vol. 149, pp. 215–225, Mar. 2019.

[40] P. Singh and N. Komodakis, "Cloud-GAN: Cloud removal for Sentinel-2 imagery using a cyclic consistent generative adversarial networks," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2018, pp. 1772–1775.

[41] J. Li et al., "Thin cloud removal in optical remote sensing images based on generative adversarial networks and physical model of cloud distortion," *ISPRS J. Photogramm. Remote Sens.*, vol. 166, pp. 373–389, Aug. 2020.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 03:30:23 UTC from IEEE Xplore. Restrictions apply.

[42] Y. Zi, F. Xie, X. Song, Z. Jiang, and H. Zhang, "Thin cloud removal for remote sensing images using a physical-model-based CycleGAN with unpaired data," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[43] W. Li, Y. Li, D. Chen, and J. C.-W. Chan, "Thin cloud removal with residual symmetrical concatenation network," *ISPRS J. Photogramm. Remote Sens.*, vol. 153, pp. 137–150, Jul. 2019.

[44] Y. Zhou, W. Jing, J. Wang, G. Chen, R. Scherer, and R. Damaševič ius, "MSAR-DefogNet: Lightweight cloud removal network for high resolution remote sensing images based on multi scale convolution," *IET Image Process.*, vol. 16, no. 3, pp. 659–668, Feb. 2022.

[45] X. Wen, Z. Pan, Y. Hu, and J. Liu, "An effective network integrating residual learning and channel attention mechanism for thin cloud removal," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[46] W. Yu, X. Zhang, and M.-O. Pun, "Cloud removal in optical remote sensing imagery using multiscale distortion-aware networks," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[47] Y. Zi, H. Ding, F. Xie, Z. Jiang, and X. Song, "Wavelet integrated convolutional neural network for thin cloud removal in remote sensing images," *Remote Sens.*, vol. 15, no. 3, p. 781, Jan. 2023.

[48] K. Enomoto et al., "Filmy cloud removal on satellite imagery with multispectral conditional generative adversarial nets," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR) Workshops*, Jul. 2017, pp. 48–56.

[49] P. Isola, J.-Y. Zhu, T. Zhou, and A. A. Efros, "Image-to-image translation with conditional adversarial networks," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jul. 2017, pp. 1125–1134.

[50] X. Wen, Z. Pan, Y. Hu, and J. Liu, "Generative adversarial learning in YUV color space for thin cloud removal on satellite imagery," *Remote Sens.*, vol. 13, no. 6, p. 1079, Mar. 2021.

[51] H. Pan, "Cloud removal for remote sensing imagery via spatial attention generative adversarial network," 2020, *arXiv:2009.13015*.

[52] M. Xu, F. Deng, S. Jia, X. Jia, and A. J. Plaza, "Attention mechanism-based generative adversarial networks for cloud removal in Landsat images," *Remote Sens. Environ.*, vol. 271, Mar. 2022, Art. no. 112902.

[53] H. Ding, Y. Zi, and F. Xie, "Uncertainty-based thin cloud removal network via conditional variational autoencoders," in *Proc. Asian Conf. Comput. Vis.*, 2022, pp. 469–485.

[54] R. Jing, F. Duan, F. Lu, M. Zhang, and W. Zhao, "Denoising diffusion probabilistic feature-based network for cloud removal in Sentinel-2 imagery," *Remote Sens.*, vol. 15, no. 9, p. 2217, Apr. 2023.

[55] P. S. Chavez Jr., "An improved dark-object subtraction technique for atmospheric scattering correction of multispectral data," *Remote Sens. Environ.*, vol. 24, no. 3, pp. 459–479, Apr. 1988.

[56] A. Makarau, R. Richter, R. Müller, and P. Reinartz, "Haze detection and removal in remotely sensed multispectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 52, no. 9, pp. 5895–5905, Sep. 2014.

[57] S. G. Narasimhan and S. K. Nayar, "Vision and the atmosphere," *Int. J. Comput. Vis.*, vol. 48, no. 3, pp. 233–254, 2002.

[58] M. Suin, K. Purohit, and A. N. Rajagopalan, "Spatially-attentive patch-hierarchical network for adaptive motion deblurring," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 3603–3612.

[59] H. Zhang, Y. Dai, H. Li, and P. Koniusz, "Deep stacked hierarchical multi-patch network for image deblurring," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 5971–5979.

[60] S. W. Zamir et al., "Multi-stage progressive image restoration," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 14816–14826.

[61] Y. Zhang, Y. Tian, Y. Kong, B. Zhong, and Y. Fu, "Residual dense network for image super-resolution," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 2472–2481.

[62] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 770–778.

[63] G. Huang, Z. Liu, L. Van Der Maaten, and K. Q. Weinberger, "Densely connected convolutional networks," in *Proc. IEEE Conf. Comput Vis. Pattern Recognit.*, Jul. 2017, pp. 4700–4708.

[64] F. Yu and V. Koltun, "Multi-scale context aggregation by dilated convolutions," 2015, *arXiv:1511.07122*.

[65] X. Yu et al., "Towards efficient and scale-robust ultra-high-definition image demoiréing," in *Proc. Eur. Conf. Comput. Vis.* Cham, Switzerland: Springer, 2022, pp. 646–662.

[66] T. Wang, K. Zhang, T. Shen, W. Luo, B. Stenger, and T. Lu, "Ultra-high-definition low-light image enhancement: A benchmark and transformer-based method," in *Proc. AAAI Conf. Artif. Intell.*, vol. 37, no. 3, 2023, pp. 2654–2662.

[67] J. Liang, J. Cao, G. Sun, K. Zhang, L. Van G., and R. Timofte, "SwinIR: Image restoration using Swin transformer," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV) Workshops*, Oct. 2021, pp. 1833–1844.

[68] Z. Liu et al., "Swin transformer: Hierarchical vision transformer using shifted windows," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 10012–10022.

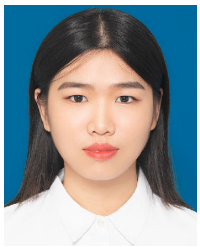
[69] J. Xu, X. Sun, Z. Zhang, G. Zhao, and J. Lin, "Understanding and improving layer normalization," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 32, 2019, pp. 4381–4391.

[70] P. Charbonnier, L. Blanc-Feraud, G. Aubert, and M. Barlaud, "Two deterministic half-quadratic regularization algorithms for computed imaging," in *Proc. 1st Int. Conf. Image Process. (ICIP)*, 1994, pp. 168–172.

[71] K. Jiang et al., "Multi-scale progressive fusion network for single image deraining," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 8346–8355.

[72] D. Lin, G. Xu, X. Wang, Y. Wang, X. Sun, and K. Fu, "A remote sensing image dataset for cloud removal," 2019, *arXiv:1901.00600*.

[73] J. Li, Z. Wu, Z. Hu, Z. Li, Y. Wang, and M. Molinier, "Deep learning based thin cloud removal fusing vegetation red edge and short wave infrared spectral information for Sentinel-2A imagery," *Remote Sens.*, vol. 13, no. 1, p. 157, Jan. 2021.



Jiao Liu received the B.S. degree in computer science and technology from Taishan Medical University, Taian, China, in 2018, and the M.S. degree in computer science and technology from the Hebei University of Technology, Tianjin, China, in 2021. She is currently pursuing the Ph.D. degree with the College of Computer Science, Nankai University, Tianjin.

Her research interests include domain adaptation and image restoration.



Bin Pan (Member, IEEE) received the B.S. and Ph.D. degrees from the School of Astronautics, Beihang University, Beijing, China, in 2013 and 2019, respectively.

Since 2019, he has been an Associate Professor with the School of Statistics and Data Science, Nankai University, Tianjin, China. He has authored or coauthored over 40 scientific articles. His research interests include cross-domain remote sensing image processing, hyperspectral image classification, unmixing, and super-resolution.

Dr. Pan serves as an Associate Editor for *Infrared Physics and Technology*. Codes for his articles are published online https://github.com/Lab-PANbin/.



Zhenwei Shi (Senior Member, IEEE) is currently a Professor and the Dean of the Image Processing Center, School of Astronautics, Beihang University, Beijing, China. He has authored or coauthored more than 200 scientific papers in refereed journals and proceedings. His research interests include remote sensing image processing and analysis, computer vision, pattern recognition, and machine learning. His personal website is http://levir.buaa.edu.cn/.

Prof. Shi serves as an Editor for IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING, *Pattern Recognition*, *ISPRS Journal of Photogrammetry and Remote Sensing*, and *Infrared Physics and Technology*.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 03:30:23 UTC from IEEE Xplore. Restrictions apply.